

6.2

najprv najdeme maticu zobrazenia uzhladom k bázi: $B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right)$

$$[f]_B^B = \left([f(v_1)]_B \mid [f(v_2)]_B \right) = \left(\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}_B \mid \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}_B \right) = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

teraz najdeme vlastné čísla matice $[f]_B^B$, ktoré sú rovnaké ako vlastné čísla operátora f

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = 0 \quad \Leftrightarrow \quad (1-\lambda)(-2-\lambda) - 4 = 0$$
$$(1-\lambda)(\lambda+3) = 0$$
$$\lambda_1 = 2$$
$$\lambda_2 = -3$$

vlastné vektory uzhladom k bázi B sú potom

$$[M_2]_B = \ker \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$[M_{-3}]_B = \ker \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

$$[M_2]_B = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} \rightarrow M_2 = \text{span} \left\{ 2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} \right\}$$

$$[M_{-3}]_B = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\} \rightarrow M_{-3} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right\}$$

~~gener~~ označme bázu $C = \left(\begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right)$

potom $\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}_C = (1, -1)$

takže

$$f^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = 1 \cdot 2^n \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} + (-1) \cdot (-3)^n \begin{pmatrix} 0 \\ 1 \\ -2 \\ -3 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n \\ 2^{n+1} - (-3)^n \\ 2^n + 2(-3)^n \\ 2^{n+2} + 3(-3)^n \end{pmatrix}$$